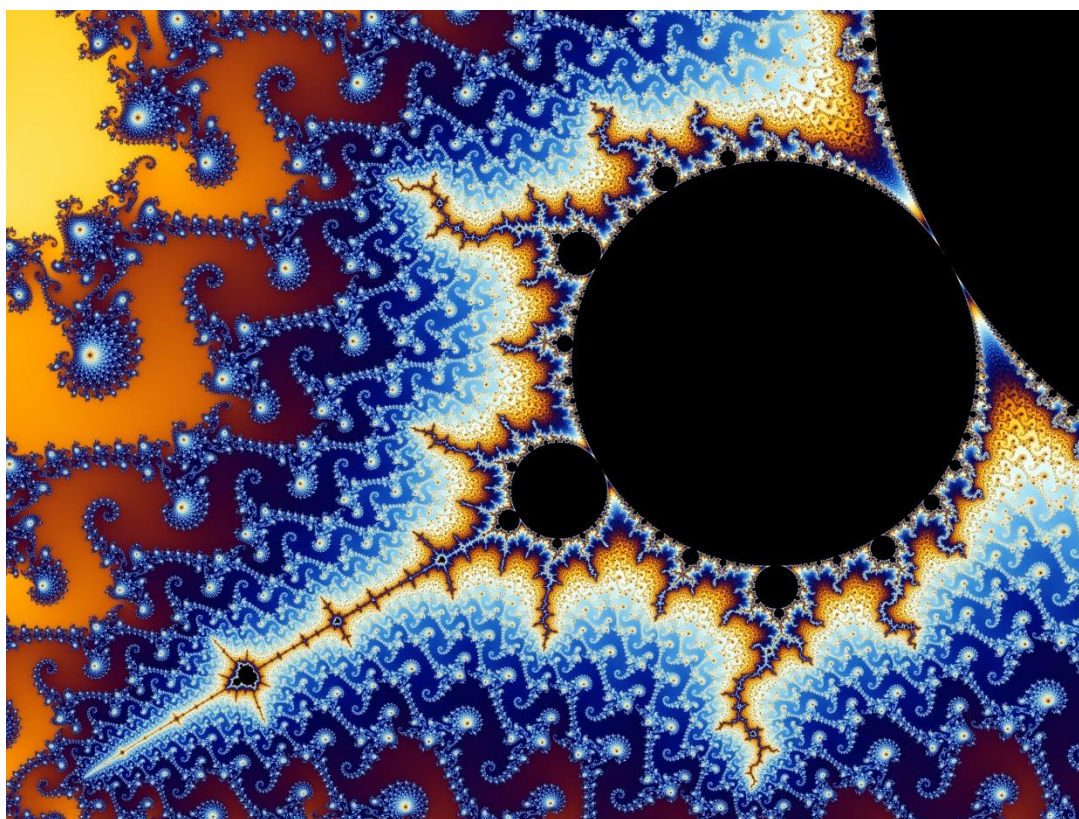




Tour de Maths 2019

Leg 1

Question Paper



The Mandelbrot set

5 Mark Questions

Question 1

Tasneem was going to buy eight sweets at the tuckshop but it turned out she was R7 short. Instead, she only buys seven sweets and is left with R5 to spare.

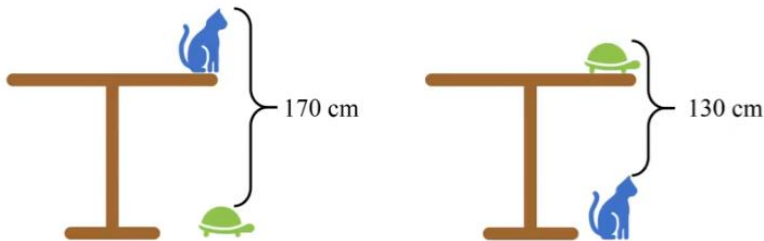
How much did a single sweet cost if all the sweets she bought cost the same?

Question 2

Which day of the week was it a million seconds ago?

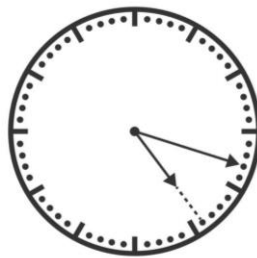
Question 3

How tall is the table?



Question 4

A clock with no numbers fell to the ground.



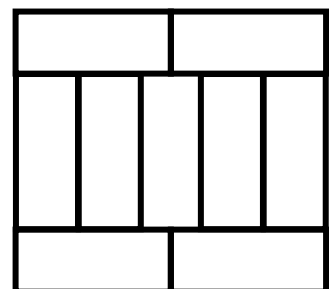
What time is it?

- (a) 4:18 (b) 5:23 (c) 10:48 (d) 11:53

Question 5

In the diagram, a large rectangle is made up of nine small identical rectangles whose longest sides are 10cm long.

What is the perimeter of the large rectangle?



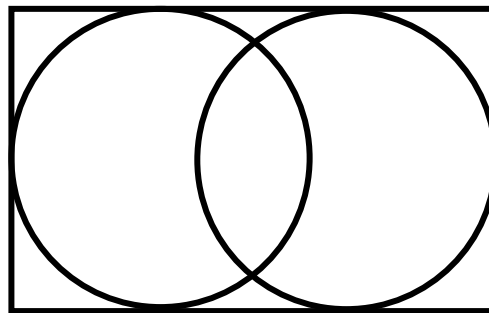
Question 6

A hotel on an island in the Caribbean advertises using the slogan “350 days of sun every year!”. According to the advert, what is the smallest number of days Rihanna has to stay at the hotel in 2019 to be certain of having two consecutive days of sun?



Question 7

The diagram shows a rectangle of dimensions 7×11 containing two circles, each touching three of the sides of the rectangle. What is the distance between the centres of the two circles?



Question 8

This is a KenKen puzzle.

To solve a 4×4 KenKen puzzle, fill in the blank squares so that each row and each column contain only all of the digits 1, 2, 3 or 4. The heavy lines indicate areas (called cages) that contain groups of numbers that can be combined (in any order) to produce the result shown in the cage, with the indicated operation. For example, $12 \times$ means you can multiply the values together to produce 12.

Example of a 3×3 KenKen puzzle:

1 -	8 +	
	3 ×	

3	2	1
2	1	3
1	3	2

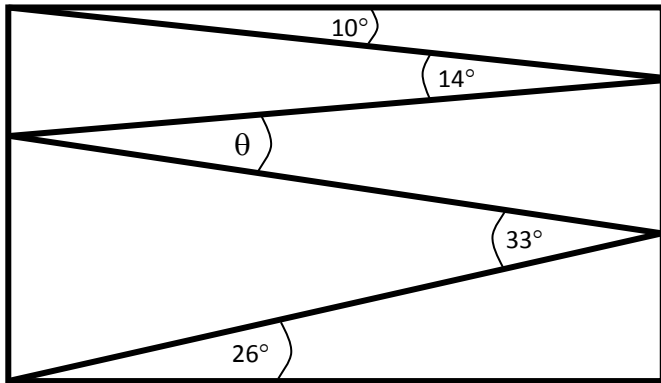
What is the sum of the numbers in the two shaded squares?

2 -		2 -	3 +
12 ×			
6 +		1 -	
		2 ÷	

Question 9

Naledi draws a zig-zag line inside a rectangle, creating angles of 10° , 14° , 33° and 26° as shown.

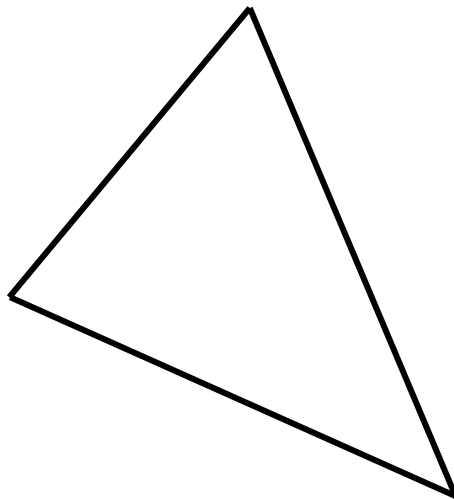
What is the size of angle θ ?



Question 10

The lengths of the sides of a triangle are x , 16 and 31, where x is the shortest side.

If the triangle is not isosceles, what is a possible value of x ?



10 Mark Questions

Question 11

Determine natural number x so that $(x^2 + 2065)$ is the square of a natural number (a perfect square).

Question 12

Four married couples: Agatha and John, Barbara and Kevin, Celine and Deon, and Daphne and Matthew were celebrating Matthew's birthday. Everybody was sitting at a round table in such a way that each lady was seated between two gentlemen and all the couples were separated. Agatha took her seat between Kevin and Matthew. Matthew sat to the right of Agatha. John was sitting next to Daphne.



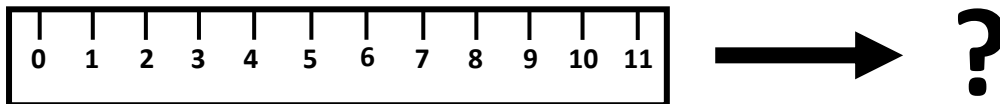
Who is sitting on the right of Barbara?

Question 13

If we erase 3 marks from a 6cm long ruler, and remove 3 numbers written below them (as in the picture), we will get a new ruler consisting of four marks only. Using this ruler, we will, however, still be able to measure in whole numbers each distance from 1 to 6. For example, we can measure 2cm because this distance lies between the remaining marks 4 and 6.



What maximum number of marks and numbers can we remove from an 11cm ruler, and yet be able to measure each distance from 1cm up to 11cm?



Fill in only the number on the answer sheet.

Question 14

Consider a 24 Hour digital clock face showing hours and minutes as below:



You place a mirror in front of the clock so that the clock face is reflected in the mirror.

How often in a 24-hour period will the reflection of the time in the mirror be the same as the actual time on the clock?

Hint: one such occasions would be at 11:11, one occasion that would not work is 23:45.



Question 15

Given: a three digit number abc . None of the digits is equal to 0.

If $abc = a! + b! + c!$ determine the number abc .

Hint:

The factorial function (symbol: !) means that we must multiply all whole numbers from our given number down to 1.

For example: $3! = 3 \times 2 \times 1$

$$8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

$$1! = 1$$

20 Mark Questions

Question 16

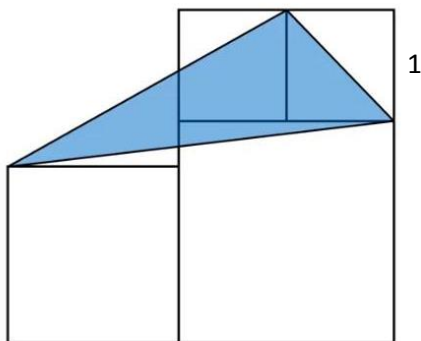
When Augustus de Morgan (a mathematician who was born and died in the 19th century) was asked about his age, he replied: I was x years old in the year x^2 .

What year was he born in?

Question 17

Consider the diagram which is made up of 4 squares, the smallest of which has a side length of 1 unit.

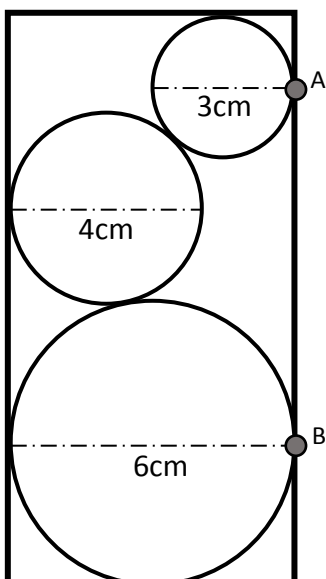
What is the area of the triangle?



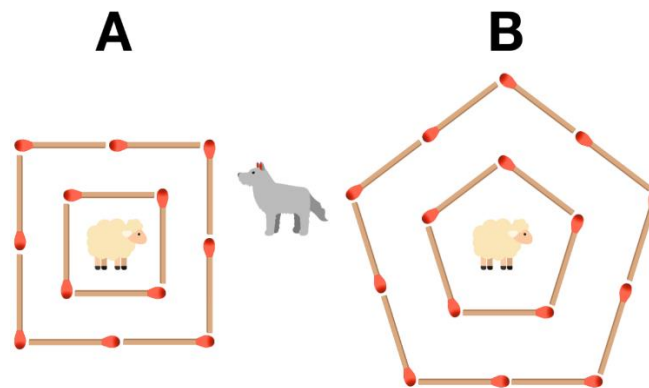
Question 18

In the diagram, three circles of diameter 3cm, 4cm and 6cm have been placed in a rectangle as shown.

What is the length of AB?



Question 19



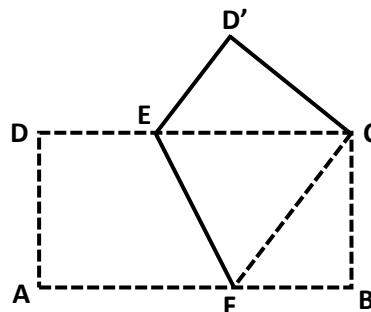
Each sheep has two fences (represented by matchsticks) protecting it from the wolf.

If any two matchsticks are removed at random, what is the probability of the wolf reaching the sheep?

Question 20

A rectangular, white sheet of paper with vertices ABCD and an area of 20cm^2 was folded and pressed in such a way that its opposite vertices A and C touched each other. In this way, pentagon BCD'E'F was created with an area of 12cm^2 ; both side were painted green, and then the paper was unfolded to regain the initial rectangle.

One side of the rectangle is now completely green and the other is two-colour. What is the area of the remaining white section?



Total: 200

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